EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB REPORT#11

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	**5**	**5**	**5**	**15**
Obtained				

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Lab Instructor.
Kaynat Rana

Experiment #11

Introduction to Binary Tree

Objective:

The objective of this lab is to understand the basic concept of Trees

- Binary Tree
- Insertion in a BT
- Deletion in a BT
- Traversing a Tree

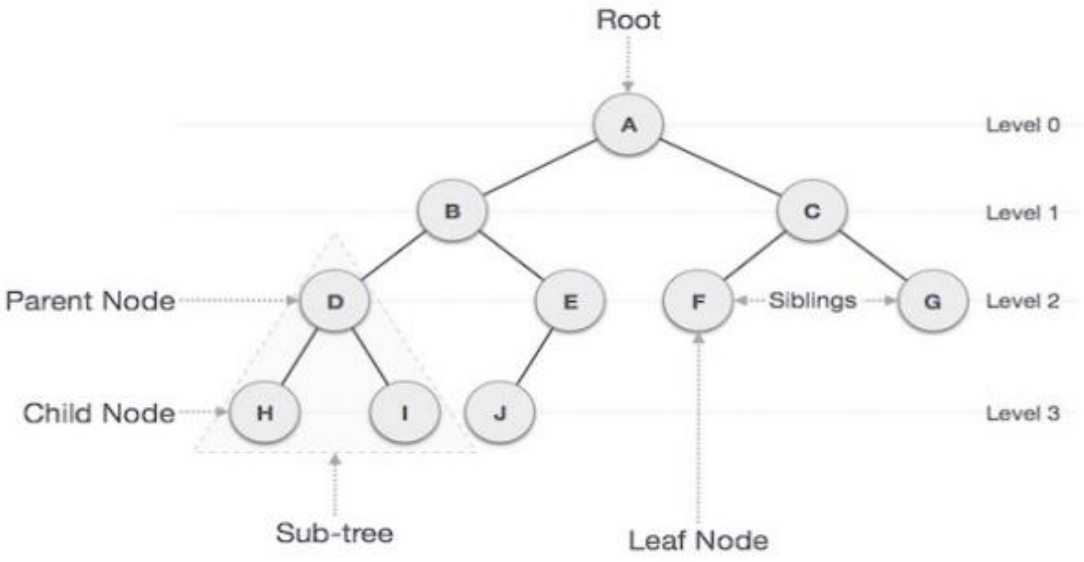
Equipment Used:

Microsoft Visual Studio

Procedure:

Trees:

Tree represents the nodes connected by edges. We will discuss binary tree or binary search tree specifically. Binary Tree is a special data structure used for data storage purposes. A binary tree has a special condition that each node can have a maximum of two children. A binary tree has the benefits of both an ordered array and a linked list as search is as quick as in a sorted array and insertion or deletion operation are as fast as in linked list.



Following are the important terms with respect to tree.

- **Path:**

Path refers to the sequence of nodes along the edges of a tree.

- **Root:**

The node at the top of the tree is called root. There is only one root per tree and one path from the root

node to any node.

- **Parent:**

Any node except the root node has one edge upward to a node called parent.

- **Child:**

The node below a given node connected by its edge downward is called its child node.

- **Leaf:**

The node which does not have any child node is called the leaf node.

- **Subtree:**

Subtree represents the descendants of a node.

- **Visiting:**

Visiting refers to checking the value of a node when control is on the node.

- **Traversing:**

Traversing means passing through nodes in a specific order.

- **Levels:**

Level of a node represents the generation of a node. If the root node is at level 0, then its next child node is at level 1, its grandchild is at level 2, and so on.

- **Keys:**

Key represents a value of a node based on which a search operation is to be carried out for a node.

BT Basic Operations:

The basic operations that can be performed on a binary search tree data structure, are the following.

- **Insert:**

Inserts an element in a tree/create a tree.

- **Search:**

Searches an element in a tree.

- **Preorder Traversal:**

Traverses a tree in a pre-order manner.

- **Inorder Traversal:**

Traverses a tree in an in-order manner.

- **Postorder Traversal:**

Traverses a tree in a post Traverses a tree in a post.

Insert Operation:

The very first insertion creates the tree. Afterwards, whenever an element is to be inserted, first locate its proper location. Start searching from the root node, then fill data from left to right in a proper manner that each node should have no more than two children (i.e., can be 0,1 or 2).

Node for a Tree Data:

```
struct nodeTree
{
char data;
nodeTree *left;
nodeTree *right;
}*root = NULL;
```

Insertion a new node:

```
nodeTree* insert(int arr[], nodeTree* root, int i, int n)
{
```

```
// Base case for recursion
if (i < n)
{
nodeTree* temp = new nodeTree;
temp->data = arr[i];
temp->left = NULL;
temp->right = NULL;
root = temp;
// insert left child
root->left = insert(arr, root->left, 2*i + 1, n);
// insert right child
root->right = insert(arr, root->right, 2*i + 2, n);
}
return root;
}
```

BT Traversal:

- **Inorder:**

The ordering is the left subtree, the current node, the right subtree.

- **Preorder:**

The ordering is the current node, the left subtree, the right subtree.

- **Post order:**

The ordering is: the left subtree, the right subtree, the current node.

Procedure:

Inorder Traversal:

Algorithm In order (tree)

1. Traverse the left subtree, i.e., call inorder(left-subtree).
2. Visit the root.
3. Traverse the right subtree, i.e., call in order(right-subtree).

Preorder Traversal:

Algorithm Preorder(tree)

1. Visit the root.
2. Traverse the left subtree, i.e., call Preorder(left-subtree).
3. Traverse the right subtree, i.e., call Preorder(right-subtree).

Uses of Preorder:

Preorder traversal is used to create a copy of the tree. Preorder traversal is also used to get prefix expression on of an expression tree.

Post order Traversal:

Algorithm Post order (tree)

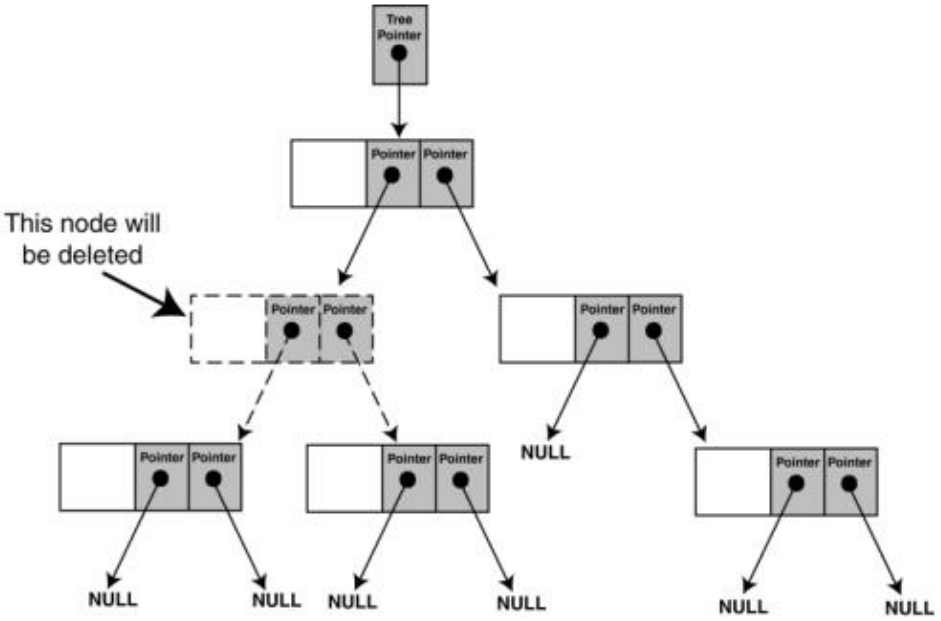
1. Traverse the left subtree, i.e., call Post order(left-subtree).
2. Traverse the right subtree, i.e., call Post order(right-subtree).
3. Visit the root.

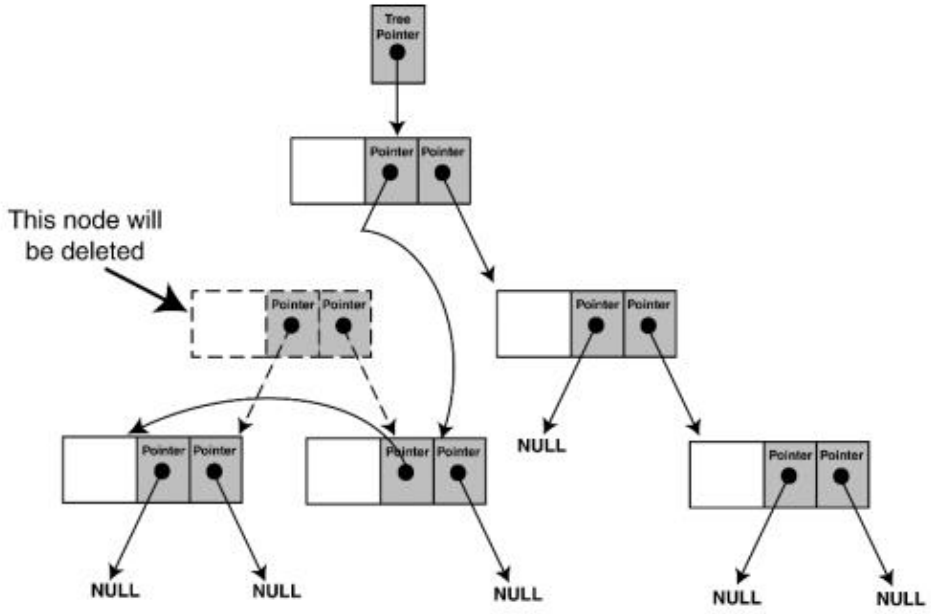
Uses of Post order:

Post order traversal is used to delete the tree. Postorder traversal is also useful to get the postfix expression of an expression tree.

BT Deletion:

While deleting a node in BT, we want to delete the node, but preserve the sub-trees, that the node links to. There are 2 possible situations to be faced when deleting a non-leaf node:
1) The node has one child.
2) The node has two children.
We cannot attach both of the node's subtrees to its parent in case 2. One solution is to:
a) find a position in the right subtree to attach the left subtree.
b) attach the node's right subtree to the parent.





Tasks

Task 1:

Write a C++ program to create a Binary Tree ADT. Your tree must be able to perform the following functions

a. Insertion in a tree
b. Inorder Traversal
c. Preorder Traversal
d. Postorder Traversal
e. Deletion of any node

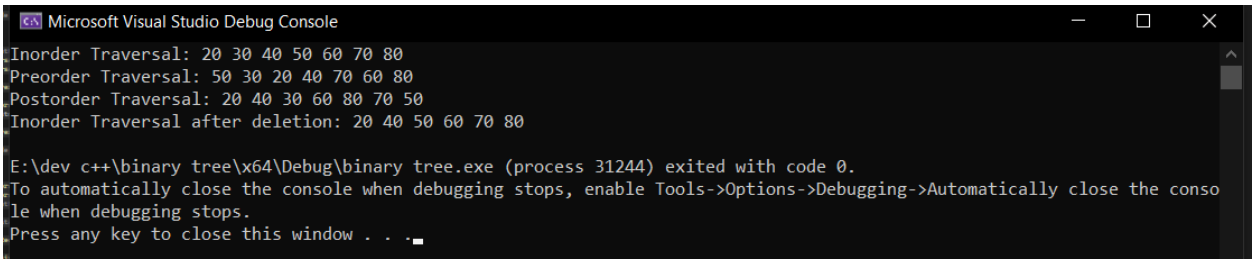
CODE:

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node* left;
    Node* right;
};
class BinaryTree
{
private:
    Node* root;
public:
    BinaryTree()
    {
        root = nullptr;
    }
    Node* createNode(int data)
    {
        Node* newNode = new Node();
        if (newNode == nullptr)
        {
            cout << "Failed to create a new node." << endl;
            return nullptr;
        }
        newNode->data = data;
        newNode->left = newNode->right = nullptr;
        return newNode;
    }
    void insert(int data)
    {
        root = insertRecursive(root, data);
    }
    Node* insertRecursive(Node* currentNode, int data)
    {
        if (currentNode == nullptr)
        {
            currentNode = createNode(data);
        }
        else if (data <= currentNode->data)
        {
            currentNode->left = insertRecursive(currentNode->left, data);
        }
        else
        {
            currentNode->right = insertRecursive(currentNode->right, data);
        }
        return currentNode;
    }
    void inorderTraversal()
```

```
{
    inorderTraversalRecursive(root);
    cout << endl;
}
void inorderTraversalRecursive(Node* currentNode)
{
    if (currentNode == nullptr)
    {
        return;
    }
    inorderTraversalRecursive(currentNode->left);
    cout << currentNode->data << " ";
    inorderTraversalRecursive(currentNode->right);
}
void preorderTraversal()
{
    preorderTraversalRecursive(root);
    cout << endl;
}
void preorderTraversalRecursive(Node* currentNode)
{
    if (currentNode == nullptr)
    {
        return;
    }
    cout << currentNode->data << " ";
    preorderTraversalRecursive(currentNode->left);
    preorderTraversalRecursive(currentNode->right);
}
void postorderTraversal()
{
    postorderTraversalRecursive(root);
    cout << endl;
}
void postorderTraversalRecursive(Node* currentNode)
{
    if (currentNode == nullptr)
    {
        return;
    }
    postorderTraversalRecursive(currentNode->left);
    postorderTraversalRecursive(currentNode->right);
    cout << currentNode->data << " ";
}
void remove(int data)
{
    root = removeRecursive(root, data);
}
Node* removeRecursive(Node* currentNode, int data)
{
    if (currentNode == nullptr)
    {
        return nullptr;
    }
    else if (data < currentNode->data)
    {
        currentNode->left = removeRecursive(currentNode->left, data);
    }
    else if (data > currentNode->data)
    {
        currentNode->right = removeRecursive(currentNode->right, data);
    }
```

```
        else
        {
            if (currentNode->left == nullptr && currentNode->right == nullptr)
            {
                delete currentNode;
                currentNode = nullptr;
            }
            else if (currentNode->left == nullptr)
            {
                Node* temp = currentNode;
                currentNode = currentNode->right;
                delete temp;
            }
            else if (currentNode->right == nullptr)
            {
                Node* temp = currentNode;
                currentNode = currentNode->left;
                delete temp;
            }
            else
            {
                Node* minValueNode = findMinValueNode(currentNode->right);
                currentNode->data = minValueNode->data;
                currentNode->right = removeRecursive(currentNode->right, minValueNode-
>data);
            }
        }
        return currentNode;
    }
    Node* findMinValueNode(Node* currentNode)
    {
        while (currentNode->left != nullptr)
        {
            currentNode = currentNode->left;
        }
        return currentNode;
    }
};
int main()
{
    BinaryTree tree;
    tree.insert(50);
    tree.insert(30);
    tree.insert(20);
    tree.insert(40);
    tree.insert(70);
    tree.insert(60);
    tree.insert(80);
    cout << "Inorder Traversal: ";
    tree.inorderTraversal();
    cout << "Preorder Traversal: ";
    tree.preorderTraversal();
    cout << "Postorder Traversal: ";
    tree.postorderTraversal();
    tree.Remove(30);
    cout << "Inorder Traversal after deletion: ";
    tree.inorderTraversal();
    return 0;
}
```

OUTPUT:



Conclusion:

In conclusion, a binary tree is a fundamental data structure in computer science and is used to represent hierarchical relationships between data. It consists of nodes, each of which has at most two children, and can be traversed in several ways, including preorder, inorder, and postorder traversal. Binary trees are used in a variety of applications, including search algorithms, sorting algorithms, and data compression. Understanding the basics of binary trees is essential for any computer scientist or programmer, as it provides a foundation for more advanced data structures and algorithms.